

Notes: Petersen (RFS, 2009)

Estimating Standard Errors in Finance Panel Data Sets: Comparing Approaches

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1 Big picture

- **Question.** How should finance researchers estimate standard errors in panel data when observations are correlated within firms, across time, or both?
- **Motivation.** Finance panel data almost never consist of independent observations. A firm can have persistent unobserved characteristics across years, and a year/month can contain common shocks across firms.
- **Literature problem.** Petersen’s survey of finance papers shows that many published regressions did not adjust standard errors for within-cluster dependence, and the remaining papers used a mixture of Fama–MacBeth, fixed effects, Newey–West, and clustered methods.
- **Core lesson.** Standard errors are correct only if the estimator is robust to the actual dependence in the data. Heteroskedasticity-robust White standard errors are not enough when residuals are correlated across observations.
- **Firm effect.** If residuals are correlated within firm over time, standard errors should be clustered by firm. OLS, White, and Fama–MacBeth standard errors are generally biased downward.
- **Time effect.** If residuals are correlated across firms within the same time period, Fama–MacBeth standard errors and time clustering can work; clustering only by firm does not solve the problem.
- **Firm and time effects.** If both dimensions matter, researchers should use two-way clustering by firm and time.
- **Key intuition.** Bias depends on the dependence in *both* the independent variable and the residual. OLS standard errors are biased when $X_{it}\varepsilon_{it}$ is correlated across observations.
- **Simulation evidence.** In simulated panels, conventional standard errors can produce many more false positives than the nominal 1% level; clustering in the right dimension restores correct rejection rates.
- **Application evidence.** In an asset-pricing application, time clustering dominates because residuals and regressors are correlated across firms within months. In a corporate-finance application, firm clustering dominates because variables and residuals are persistent within firms.
- **Takeaway.** Compare White, firm-clustered, time-clustered, and two-way-clustered standard errors. Large differences diagnose the correlation structure and indicate which inference method is needed.

2 Data, simulation design, and measurement

2.1 Survey of empirical finance practice

- Petersen surveys papers in the *Journal of Finance*, *Journal of Financial Economics*, and *Review of Financial Studies* from 2001–2004.
- The survey focuses on papers with panel regressions: multiple observations on a unit such as firm, industry, country, or year.
- A large share of papers either used OLS/White standard errors or did not clearly correct for dependence in residuals.
- Among papers that did correct, common approaches included Fama–MacBeth, fixed effects, Newey–West-type adjustments, and clustered standard errors.

Interpretation: The paper is written as practical econometric guidance for applied finance. It does not only prove formulas; it explains why common empirical choices succeed or fail.

2.2 Baseline simulated panel

The basic regression is

$$Y_{it} = X_{it}\beta + \varepsilon_{it}, \quad (1)$$

where i indexes firms and t indexes years. The true slope is set to $\beta = 1$ in the simulations.

- Baseline simulated panels usually contain $N = 500$ firms and $T = 10$ years, so there are 5,000 observations.
- The standard deviation of X is normalized to one.
- The standard deviation of the residual is set to two.
- Simulations are repeated 5,000 times.
- The paper records the average estimated coefficient, the true standard deviation of estimated coefficients, the average estimated standard error, and the fraction of $|t| > 2.58$ rejections.

Interpretation: The key diagnostic is whether the reported standard error matches the simulated sampling variation of the coefficient estimate. Correct standard errors should also generate about 1% rejections at the 1% two-sided level.

2.3 Fixed firm effect

The residual and regressor can contain firm components:

$$\varepsilon_{it} = \gamma_i + \eta_{it}, \quad (2)$$

$$X_{it} = \mu_i + v_{it}. \quad (3)$$

The within-firm correlations are

$$\text{corr}(X_{it}, X_{is}) = \rho_X, \quad s \neq t, \quad (4)$$

$$\text{corr}(\varepsilon_{it}, \varepsilon_{is}) = \rho_\varepsilon, \quad s \neq t. \quad (5)$$

Interpretation: ρ_X measures how persistent the regressor is within firm; ρ_ε measures how persistent the residual is within firm. Bias appears when both matter.

2.4 Fixed time effect

The residual and regressor can instead contain common time components:

$$\varepsilon_{it} = \delta_t + \eta_{it}, \quad (6)$$

$$X_{it} = \zeta_t + v_{it}. \quad (7)$$

Interpretation: A time effect creates cross-sectional dependence within a year or month. This is common in asset pricing when many firms are exposed to common shocks.

2.5 Temporary firm effect

The paper later allows the firm effect to decay:

$$\eta_{it} = \phi\eta_{i,t-1} + \sqrt{1 - \phi^2}\zeta_{it}, \quad (8)$$

so within-firm correlation depends on the time lag.

Interpretation: A persistent but nonfixed firm component is more realistic than a pure fixed effect. The simulations show that clustering by firm remains robust even when the correlation structure is not exactly fixed.

2.6 Empirical applications

- **Asset-pricing application.** The paper reproduces a Daniel–Titman-style regression of stock returns on characteristics such as book-to-market, book return, market return, and share issuance.
- **Corporate-finance application.** The paper estimates a leverage regression using Compustat data from 1965–2003, with market debt ratio as the dependent variable.
- These two applications are used to show how actual finance data can have very different dependence structures.

Interpretation: The applications translate the simulation advice into a practical diagnostic: compare standard errors under different clustering schemes to infer whether firm or time dependence is important.

3 Models and empirical specifications

3.1 OLS coefficient and standard error

The OLS coefficient can be written as

$$\hat{\beta}_{OLS} = \beta + \frac{\sum_i \sum_t X_{it} \varepsilon_{it}}{\sum_i \sum_t X_{it}^2}. \quad (9)$$

If all observations are independent, the asymptotic variance is

$$\text{Avar}(\hat{\beta}_{OLS} - \beta) = \frac{\sigma_\varepsilon^2}{\sigma_X^2 NT}. \quad (10)$$

Interpretation: Standard OLS treats each firm-year as new independent information. That assumption is exactly what fails in panels with within-firm or within-time dependence.

3.2 Bias from a fixed firm effect

With fixed firm effects in both X and ε , the true asymptotic variance becomes

$$\text{Avar}(\hat{\beta}_{OLS} - \beta) = \frac{\sigma_\varepsilon^2}{\sigma_X^2 NT} [1 + (T - 1)\rho_X \rho_\varepsilon]. \quad (11)$$

Interpretation: If $\rho_X \rho_\varepsilon > 0$, the OLS standard error is too small. Additional years do not provide as much independent information as OLS assumes.

3.3 Clustered standard errors

Firm-clustered standard errors estimate the variance by summing $X_{it} \hat{\varepsilon}_{it}$ within each firm before squaring:

$$\hat{V}_{cluster(F)}(\hat{\beta}) \propto \sum_i \left(\sum_t X_{it} \hat{\varepsilon}_{it} \right)^2. \quad (12)$$

Interpretation: Clustering by firm allows arbitrary correlation across observations of the same firm. It does not require estimating the exact correlation pattern.

3.4 Fama–MacBeth standard errors

Fama–MacBeth estimates a separate cross-sectional regression each period and uses the time-series variation of the period-by-period coefficients:

$$\hat{\beta}_{FM} = \frac{1}{T} \sum_{t=1}^T \hat{\beta}_t, \quad SE_{FM} = \sqrt{\frac{1}{T(T-1)} \sum_t (\hat{\beta}_t - \hat{\beta}_{FM})^2}. \quad (13)$$

Interpretation: Fama–MacBeth is designed for cross-sectional dependence within time periods, not for persistent firm-level dependence over time.

3.5 Two-way clustering

When both firm and time dependence matter, two-way clustering combines firm and time clusters and subtracts the overlap:

$$\hat{V}_{F,T} = \hat{V}_F + \hat{V}_T - \hat{V}_{White}. \quad (14)$$

Interpretation: This is the practical rule for data with both within-firm serial correlation and same-period cross-sectional correlation.

3.6 Fixed effects, GLS, and adjusted Fama–MacBeth

- Firm fixed effects remove fixed firm components from the regression, but they do not automatically solve serial correlation if the remaining firm component is persistent.
- GLS can improve efficiency if the assumed covariance structure is correct, but standard errors still need to be robust to the remaining dependence.
- Adjusted Fama–MacBeth standard errors can correct first-order autocorrelation in coefficient estimates, but they can still fail under more complex firm dependence.

Interpretation: Modeling the mean structure is not a substitute for robust inference. Fixed effects and GLS can help, but clustering is still often needed.

4 Identification and empirical design

4.1 What the simulations identify

- The simulations know the true data-generating process and true coefficient.
- This lets the paper compare the reported standard error with the actual sampling standard deviation of the coefficient estimate.
- The simulation varies the strength of dependence in the regressor and residual separately.
- Rejection rates show the practical cost of using the wrong standard error.

Interpretation: Since the true standard error is known in simulation, the paper can directly measure bias and false positives.

4.2 Why the product $\rho_X \rho_\varepsilon$ matters

- Residual correlation alone is not enough to create bias in the OLS standard error.
- The regressor must also be correlated across the same observations.
- Standard error bias reflects covariance in $X_{it}\varepsilon_{it}$, not only covariance in ε_{it} .

Interpretation: This explains why finance variables with persistent regressors and persistent residuals are especially vulnerable.

4.3 How to diagnose real data

- If firm-clustered standard errors are much larger than White standard errors, there is likely an important firm effect.
- If time-clustered standard errors are much larger than White standard errors, there is likely an important time effect.
- If two-way clustered standard errors exceed both one-way clustered estimates, both dimensions likely matter.
- Autocorrelation plots of regressors and residuals help explain why the standard errors differ.

Interpretation: The paper turns standard-error comparisons into a diagnostic tool for model misspecification.

4.4 Limits of the exercise

- The simulations are stylized and focus on dependence rather than endogeneity.
- Clustered standard errors require enough clusters; few clusters can still produce biased or noisy inference.
- The paper is about inference conditional on a regression specification, not about identifying causal effects.
- In real applications, researchers should also think about omitted variables, fixed effects, and research design.

Interpretation: Correct standard errors do not make a bad design causal, but they are necessary for valid inference in any design.

5 Tables and figures

The figures and tables below are directly cropped from the paper and grouped compactly.

5.1 Figure 1 and Table 1: fixed firm effects

Read:

- Figure 1 illustrates a residual cross-product matrix when observations are correlated within firm but independent across firms.
- Table 1 shows that OLS standard errors are correct only when either the regressor or residual has no firm effect.
- When both X and ε have firm effects, OLS standard errors are too small and rejection rates are too high.
- Firm-clustered standard errors match the true sampling variability.

Economic message: A fixed firm effect creates repeated information within firm. Clustering by firm fixes the inference problem because it treats each firm as the independent unit.

	Firm 1			Firm 2			Firm 3		
Firm 1	ε_{11}^2	$\varepsilon_{11} \varepsilon_{12}$	$\varepsilon_{11} \varepsilon_{13}$	0	0	0	0	0	0
	$\varepsilon_{12} \varepsilon_{11}$	ε_{12}^2	$\varepsilon_{12} \varepsilon_{13}$	0	0	0	0	0	0
	$\varepsilon_{13} \varepsilon_{11}$	$\varepsilon_{13} \varepsilon_{12}$	ε_{13}^2	0	0	0	0	0	0
Firm 2	0	0	0	ε_{21}^2	$\varepsilon_{21} \varepsilon_{22}$	$\varepsilon_{21} \varepsilon_{23}$	0	0	0
	0	0	0	$\varepsilon_{22} \varepsilon_{21}$	ε_{22}^2	$\varepsilon_{22} \varepsilon_{23}$	0	0	0
	0	0	0	$\varepsilon_{23} \varepsilon_{21}$	$\varepsilon_{23} \varepsilon_{22}$	ε_{23}^2	0	0	0
Firm 3	0	0	0	0	0	0	ε_{31}^2	$\varepsilon_{31} \varepsilon_{32}$	$\varepsilon_{31} \varepsilon_{33}$
	0	0	0	0	0	0	$\varepsilon_{32} \varepsilon_{31}$	ε_{32}^2	$\varepsilon_{32} \varepsilon_{33}$
	0	0	0	0	0	0	$\varepsilon_{33} \varepsilon_{31}$	$\varepsilon_{33} \varepsilon_{32}$	ε_{33}^2

Figure 1
Residual cross product matrix: Assumptions about zero covariances

The figure shows a sample covariance matrix of the residuals. Assumptions about the elements of this matrix and which are zero is the source of difference in the various standard error estimates. The standard OLS assumption is that only the diagonal terms are nonzero. Standard errors clustered by firm assume that the correlation of the residuals within the cluster may be nonzero (these elements are shaded). This cluster assumption assumes that residuals across clusters are uncorrelated. These are recorded as zero in the matrix.

Figure 1: within-firm covariance structure.

		Source of independent variable volatility			
		0%	25%	50%	75%
Avg(β_{OLS})					
Std(β_{OLS})					
Avg(SE_{OLS})					
% Sig(T_{OLS})					
Avg(SE_c)					
Avg(SE_c)					
Source of residual volatility	0%	1.0004	1.0006	1.0002	1.0001
		0.0286	0.0288	0.0279	0.0283
		0.0283	0.0283	0.0283	0.0283
		[0.0098]	[0.0088]	[0.0094]	[0.0094]
25%		0.0283	0.0282	0.0282	0.0282
		[0.0108]	[0.0092]	[0.0096]	[0.0098]
		1.0004	0.9997	0.9999	0.9997
		0.0287	0.0353	0.0403	0.0468
50%		0.0283	0.0283	0.0283	0.0283
		[0.0116]	[0.0348]	[0.0678]	[0.1174]
		0.0283	0.0353	0.0411	0.0463
		[0.0120]	[0.0064]	[0.0112]	[0.0118]
75%		1.0001	1.0002	1.0007	0.9993
		0.0289	0.0414	0.0508	0.0577
		0.0283	0.0283	0.0283	0.0283
		[0.0124]	[0.0770]	[0.1544]	[0.2076]
		1.0000	1.0004	0.9995	1.0016
		0.0285	0.0459	0.0594	0.0698
		0.0283	0.0283	0.0283	0.0283
		[0.0128]	[0.1090]	[0.2230]	[0.2906]
		0.0282	0.0462	0.0589	0.0693
		[0.0128]	[0.0114]	[0.0094]	[0.0112]

The table contains estimates of the coefficient and standard errors based on five thousand simulated panel data sets, each of which contains five hundred firms and ten years per firm. The true slope coefficient is 1, the standard deviation of the independent variable is 1, and the standard deviation of the error term is 2 (see Equation (1)). The independent variable X and the residual are specified as

$$X_{it} = \beta_1 + \eta_{it}$$

$$\varepsilon_{it} = \eta_{it} + \eta_{it}$$

The fraction of X 's variance, which is due to a firm-specific component ($\text{Var}(\eta_{it})/\text{Var}(X)$), varies across the columns of the table from 0% (no firm effect) to 75% and the fraction of the residual variance, which is due to a firm-specific component ($\text{Var}(\eta_{it})/\text{Var}(\varepsilon)$), varies across the rows of the table from 0% (no firm effect) to 75%. Each cell contains the average slope coefficient estimated by OLS and the standard deviation of this estimate. This is the true standard error of the estimated coefficient. The third entry is the average standard error estimated by OLS. The percentage of OLS t -statistics that are significant at the 1% level (e.g., $|t| > 2.58$) is reported in square brackets. The fifth entry is the average standard error clustered by firm (i.e., accounts for the possible correlation between observations of the same firm in different years). The percentage of clustered t -statistics that are significant at the 1% level is reported in square brackets. For example, when 50% of the variability in both the residual and the independent variable is due to the fixed firm effect ($\rho = \rho_0 = 0.50$), the true standard error of the OLS coefficient is 0.0508. The OLS standard error estimate is 0.0283 and the clustered standard error is 0.0508. 15.3% of the OLS t -statistics are greater than 2.58 in absolute value (only 1% should be), while 0.9% of the clustered t -statistics are greater than 2.58 in absolute value.

Table 1: fixed firm effect, OLS vs. firm clustering.

5.2 Figure 2 and Figure 3: false positives and panel length

Read:

- Figure 2 compares theoretical t distributions with simulated t statistics when 50% of the variation in both X and the residual is due to a fixed firm effect.
- OLS and Fama–MacBeth t statistics are too dispersed, producing too many large absolute t statistics.
- Firm-clustered t statistics line up with the correct distribution.
- Figure 3 shows that OLS bias worsens with more years per firm when the firm effect is fixed.
- Firm clustering remains approximately unbiased as T increases.

Economic message: More observations per firm do not automatically mean more independent information. If the firm effect is fixed, adding years can make conventional standard errors look even more precise while true precision does not improve as much.

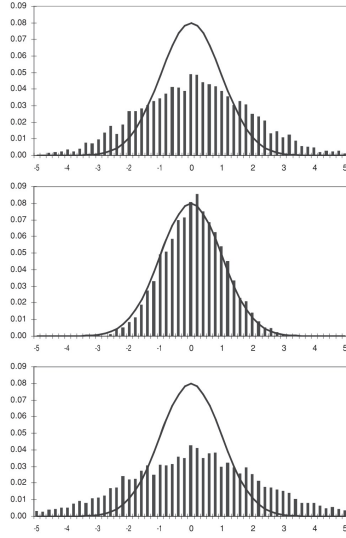


Figure 2
Distribution of simulated t -statistics
The figures contain the theoretical t -distribution (the line), and the distribution of t -statistics produced by the simulation (the bars) when 50% of the variability in the independent variable and the residual is due to a fixed firm effect. The top figure is the distribution of the t -statistics based on the OLS standard errors, the middle figure is the distribution of t -statistics based on the standard errors clustered by firm, and the bottom figure is the distribution of t -statistics based on Fama-MacBeth standard errors. When the standard error estimates are too small (as with OLS and Fama-MacBeth), there are too many t -statistics that are large in absolute value.

Figure 2: distribution of simulated t -statistics.

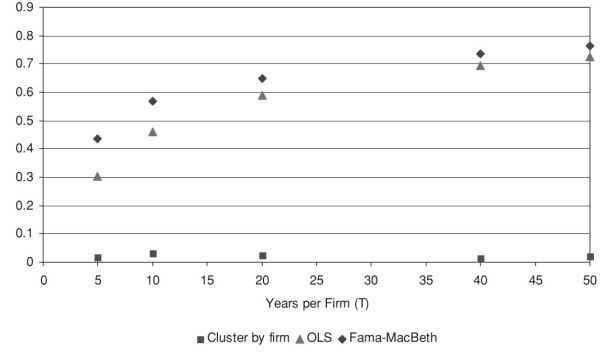


Figure 3
Bias in estimated standard errors as a function of years per cluster
The figure graphs the percentage by which OLS (triangles), clustered by firm (squares) and Fama-MacBeth (diamonds) standard errors, underestimate the true standard error in the presence of a fixed firm effect. The results are based on five thousand simulations of a data set with five thousand observations. The number of years per firm ranges from 5 to 50. The firm effect is assumed to comprise 50% of the variability in both the independent variable and the residual. The underestimates are calculated as one minus the average estimated standard error divided by the true standard deviation of the coefficient estimate. For example, the standard deviation of the coefficient estimate was 0.0406 in the simulation with five years of data ($T = 5$). The average of the OLS estimated standard errors is 0.0283. Thus the OLS underestimated the true standard error by 30% ($1 - 0.0283/0.0406$).

Figure 3: standard-error bias as T increases.

5.3 Table 2 and Figure 4: Fama–MacBeth and Newey–West under firm effects

Read:

- Table 2 shows that Fama–MacBeth standard errors are biased under firm effects.
- The bias grows as the fixed firm component in X and ε becomes larger.
- Figure 4 shows that Newey–West standard errors adapted to panel data rise with the assumed lag length but still do not match firm clustering well in this setting.

Economic message: Fama–MacBeth is not a general solution to panel dependence. It is appropriate for time effects, not persistent firm effects.

Table 2
Estimating standard errors with a firm effect Fama-MacBeth standard errors

		Source of independent variable volatility			
		0%	25%	50%	75%
Avg(β_{RM})					
Std(β_{RM})					
Avg(SE_{RM})					
% Sig(T_{RM})					
Source of residual volatility	0%	1.0004	1.0006	1.0002	1.0001
		0.0287	0.0288	0.0280	0.0283
		0.0276	0.0276	0.0277	0.0275
		[0.0288]	[0.0304]	[0.0236]	[0.0294]
	25%	1.0004	0.9997	0.9998	0.9997
		0.0288	0.0354	0.0403	0.0469
		0.0275	0.0268	0.0259	0.0250
		[0.0336]	[0.0758]	[0.1202]	[0.1918]
	50%	1.0000	1.0002	1.0007	0.9993
		0.0289	0.0415	0.0509	0.0578
		0.0276	0.0259	0.0238	0.0219
		[0.0330]	[0.1264]	[0.2460]	[0.3388]
	75%	1.0000	1.0004	0.9995	1.0016
		0.0286	0.0460	0.0595	0.0699
		0.0277	0.0248	0.0218	0.0183
		[0.0310]	[0.1778]	[0.3654]	[0.4994]

The table contains estimates of the coefficient and standard errors based on the same simulated panel data sets that are used in Table 1. Each data set contains five hundred firms and ten years per firm. The true slope coefficient is 1, the standard deviation of the independent variable is 1, and the standard deviation of the error term is 2 (see Equation (1)). The independent variable X and the residual are specified as

$$X_{it} = \mu_i + v_{it}$$

$$e_{it} = \gamma_i + \eta_{it}.$$

The fraction of X 's variance, which is due to a firm-specific component $[\text{Var}(\mu)/\text{Var}(X)]$, varies across the columns of the table from 0% (no firm effect) to 75% and the fraction of the residual variance, which is due to a firm-specific component $[\text{Var}(\gamma)/\text{Var}(e)]$, varies across the rows of the table from 0% (no firm effect) to 75%. The first entry is the average slope coefficient based on a Fama-MacBeth estimation (e.g., a regression is run for each of the ten years and the estimate is the average of the ten estimated slope coefficients). The second entry is the standard deviation of the coefficient estimated by Fama-MacBeth. This is the true standard error of the Fama-MacBeth coefficient. The third entry is the average standard error estimated by Fama-MacBeth (see Equation (9)). The percentage of Fama-MacBeth t -statistics that are significant at the 1% level (e.g., $|t| > 2.58$) is reported in square brackets. For example, when 50% of the variability in both the residual and the independent variable is due to the firm effect ($\rho_X = \rho_e = 0.50$), the true standard error of the Fama-MacBeth coefficient is 0.0509. The Fama-MacBeth standard error estimate is 0.0238, and 24.6% of t -statistics are greater than 2.58 in absolute value.

Table 2: firm effect with Fama-MacBeth standard errors.

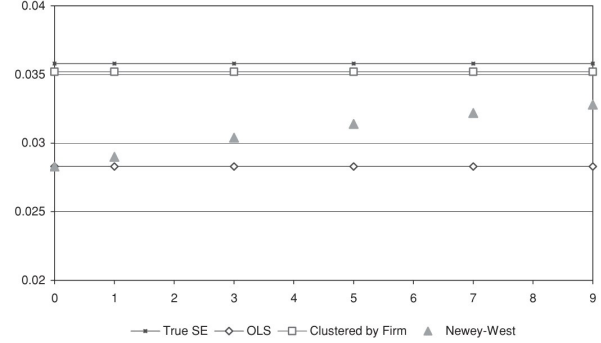


Figure 4
Relative performance of OLS, clustered, and Newey-West standard errors

The figure contains OLS standard errors, standard errors clustered by firm, and Newey-West standard errors, as well as the true standard error. Estimates are based on five thousand simulated data sets. Each data set contains five thousand observations (five hundred firms and ten years per firm). In each simulation, 25% of the variability in both the independent variable and the residual is due to a firm effect (i.e., $\rho_X = \rho_e = 0.25$). The true standard error (shaded squares), the OLS standard error estimates (empty diamonds), and the clustered standard errors (empty squares) are plotted as straight lines as they do not depend upon the assumed lag length. The Newey-West standard error estimates, which rise with the assumed lag length, are plotted as triangles.

Figure 4: OLS, firm clustering, and Newey-West.

5.4 Table 3, Figure 5, and Table 4: fixed time effects

Read:

- Table 3 shows the mirror case: residuals and regressors are correlated within time periods across firms.
- OLS standard errors are too small when the time effect appears in both X and ε .
- Time-clustered standard errors correct the problem, but their quality depends on having enough time clusters.
- Figure 5 shows that with few time clusters, clustered standard errors can still be biased/noisy; performance improves as the number of clusters rises.
- Table 4 shows that Fama-MacBeth standard errors work well in the pure time-effect case.

Economic message: The relevant cluster dimension depends on the economic source of dependence. Equity-return regressions often need time clustering because common shocks hit many firms simultaneously.

Table 3
Estimating standard errors with a time effect: OLS and clustered standard errors

		Source of independent variable volatility			
		0%	25%	50%	75%
		Avg(β_{OLS})			
		Std(β_{OLS})			
		Avg(SE_{OLS})			
		% Sig(T_{OLS})			
		Avg(SE_C)			
		% Sig(T_C)			
Source of residual volatility	0%	1.0004	1.0002	1.0006	0.9994
		0.0286	0.0291	0.0293	0.0314
		0.0283	0.0288	0.0295	0.0306
		[0.0098]	[0.0094]	[0.0088]	[0.0114]
		0.0277	0.0276	0.0275	0.0270
		[0.0330]	[0.0304]	[0.0348]	[0.0520]
	25%	1.0006	1.0043	0.9962	0.9996
		0.0284	0.1490	0.2148	0.2874
		0.0279	0.0284	0.0289	0.0300
		[0.0114]	[0.6064]	[0.7270]	[0.7874]
		0.0268	0.1297	0.1812	0.2305
		[0.0320]	[0.0360]	[0.0506]	[0.0736]
	50%	0.9996	0.9997	0.9919	1.0079
		0.0276	0.2138	0.3015	0.3986
		0.0274	0.0278	0.0282	0.0292
		[0.0100]	[0.7298]	[0.8096]	[0.8536]
		0.0258	0.1812	0.2546	0.3248
		[0.0294]	[0.0458]	[0.0596]	[0.0756]
	75%	1.0002	0.9963	0.9970	0.9908
		0.0273	0.2620	0.3816	0.4927
		0.0267	0.0271	0.0276	0.0284
		[0.0110]	[0.7994]	[0.8586]	[0.8790]
		0.0244	0.2215	0.3141	0.3986
		[0.0322]	[0.0402]	[0.0588]	[0.0768]

The table contains estimates of the coefficient and standard errors based on five thousand simulated panel data sets, each of which contains five hundred firms and ten years per firm. The true slope coefficient is 1, the standard deviation of the independent variable is 1, and the standard deviation of the error term is 2 (see Equation (1)). The independent variable X and the residual are specified as

$$X_{it} = \zeta_t + v_{it}$$

$$\varepsilon_{it} = \delta_t + \eta_{it}$$

The fraction of X 's variance, which is due to a time-specific component $[\text{Var}(\zeta)/\text{Var}(X)]$, varies across the columns of the table from 0% (no time effect) to 75% and the fraction of the residual variance, which is due to a time-specific component $[\text{Var}(\delta)/\text{Var}(\varepsilon)]$, varies across the rows of the table from 0% (no time effect) to 75%. Each cell contains the average estimated slope coefficient from OLS and the standard deviation of this estimate. This is the true standard error of the estimated coefficient. The third entry is the average standard error estimated by OLS. The percentage of OLS t -statistics that are significant at the 1% level (e.g., $|t| > 2.58$) is reported in square brackets. The fifth entry is the average standard error clustered by year (i.e., accounts for the possible correlation between observations on different firms in the same year). The percentage of clustered t -statistics that are significant at the 1% level (e.g., $|t| > 2.58$) is reported in square brackets.

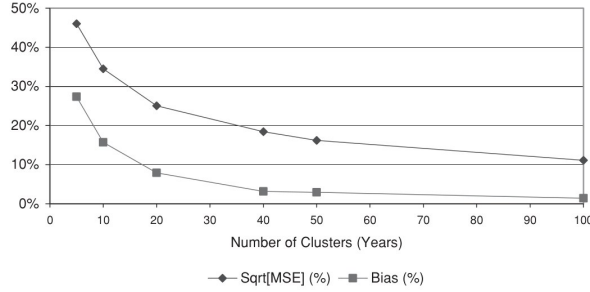


Figure 5
True standard errors and clustered standard errors as a function of cluster size (T)
The figure graphs the bias (squares) and mean squared error (MSE) (diamonds) as a function of the number of years (clusters) used in each simulation. The bias is the estimated clustered standard error minus the true standard error. The MSE is the average value of the squared difference between the estimated standard error and the true standard error.

$$MSE = E[(\bar{SE} - SE_{true})^2]$$

$$= E[(\bar{SE} - \bar{SE})^2 + (\bar{SE} - SE_{true})^2]$$

$$= \text{Var}(\bar{SE}) + \text{Bias}(\bar{SE})^2.$$

The MSE is equal to the variance of the standard error plus the bias squared. Both the bias and the MSE are divided by the true standard error and thus are expressed as a percentage. Each simulated data set has five thousand observations. In each simulation, 25% of the variability in both the independent variable and the residual is due to the time effect (i.e., $\rho_X = \rho_\varepsilon = 0.25$). The standard errors are averaged across five thousand simulations. In these simulations, underestimation of the standard errors ranges from 27% when there are five years in the simulated data set to 1% when there are one hundred years in the simulated data set.

Figure 5: cluster count, bias, and MSE.

Table 4
Estimating standard errors with a time effect: Fama-MacBeth standard errors

		Source of independent variable volatility			
		0%	25%	50%	75%
		Avg(β_{FM})			
		Std(β_{FM})			
		Avg(SE_{FM})			
		% Sig(T_{FM})			
Source of residual volatility	0%	1.0004	1.0004	1.0007	0.9991
		0.0287	0.0331	0.0396	0.0573
		0.0278	0.0318	0.0390	0.0553
		[0.0310]	[0.0312]	[0.0252]	[0.0338]
	25%	1.0005	1.0003	1.0006	0.9999
		0.0252	0.0284	0.0343	0.0496
		0.0239	0.0276	0.0338	0.0480
		[0.0376]	[0.0296]	[0.0284]	[0.0294]
	50%	1.0000	1.0002	1.0006	1.0007
		0.0200	0.0231	0.0280	0.0394
		0.0195	0.0227	0.0276	0.0387
		[0.0254]	[0.0304]	[0.0272]	[0.0278]
	75%	1.0001	0.9996	1.0000	0.9999
		0.0142	0.0161	0.0200	0.0285
		0.0138	0.0159	0.0196	0.0276
		[0.0308]	[0.0302]	[0.0284]	[0.0300]

The table contains estimates of the coefficient and standard errors based on the same five thousand simulated panel data sets that are used in Table 3. Each data set contains five hundred firms and ten years per firm. The true slope coefficient is 1, the standard deviation of the independent variable is 1, and the standard deviation of the error term is 2 (see Equation (1)). The independent variable X and the residual are specified as

$$X_{it} = \zeta_t + v_{it}$$

$$\varepsilon_{it} = \delta_t + \eta_{it}$$

The fraction of X 's variance, which is due to a time-specific component $[\text{Var}(\zeta)/\text{Var}(X)]$, varies across the columns of the table from 0% (no time effect) to 75% and the fraction of the residual variance, which is due to a time-specific component $[\text{Var}(\delta)/\text{Var}(\varepsilon)]$, varies across the rows of the table from 0% (no time effect) to 75%. The first entry is the average slope coefficient based on a Fama-MacBeth estimation (e.g., the regression is run for each of the ten years and the estimate is the average of the ten estimated slope coefficients). The second entry is the standard deviation of the coefficient estimated by Fama-MacBeth. This is the true standard error of the Fama-MacBeth coefficient. The third entry is the average standard error estimated by the Fama-MacBeth procedure (e.g., Equation (9)). The percentage of Fama-MacBeth t -statistics that are significant at the 1% level (e.g., $|t| > 2.58$) is reported in square brackets.

Table 4: time effect with Fama-MacBeth standard errors.

5.5 Figure 6 and Figure 7: two-way dependence

Read:

- Figure 6 shows a residual cross-product matrix with both firm and time effects.

- Nonzero covariance appears within firms across years and within years across firms.
- Figure 7 compares rejection rates when standard errors are clustered by firm, by time, or by both.
- One-way clustering can fail when the other dimension has few clusters or strong dependence.
- Two-way clustering is most reliable when both dimensions matter and there are enough clusters in both dimensions.

Economic message: If there are two sources of dependence, clustering by only one dimension can leave the other source unaddressed. Two-way clustering combines both corrections.

	Firm 1			Firm 2			Firm 3		
Firm 1	ϵ_{11}^2	$\epsilon_{11} \epsilon_{12}$	$\epsilon_{11} \epsilon_{13}$	$\epsilon_{11} \epsilon_{21}$	0	0	$\epsilon_{11} \epsilon_{31}$	0	0
	$\epsilon_{12} \epsilon_{11}$	ϵ_{12}^2	$\epsilon_{12} \epsilon_{13}$	0	$\epsilon_{12} \epsilon_{22}$	0	0	$\epsilon_{12} \epsilon_{32}$	0
	$\epsilon_{13} \epsilon_{11}$	$\epsilon_{13} \epsilon_{12}$	ϵ_{13}^2	0	0	$\epsilon_{13} \epsilon_{23}$	0	0	$\epsilon_{13} \epsilon_{33}$
Firm 2	$\epsilon_{21} \epsilon_{11}$	0	0	ϵ_{21}^2	$\epsilon_{21} \epsilon_{22}$	$\epsilon_{21} \epsilon_{23}$	$\epsilon_{21} \epsilon_{31}$	0	0
	0	$\epsilon_{22} \epsilon_{12}$	0	$\epsilon_{22} \epsilon_{21}$	ϵ_{22}^2	$\epsilon_{22} \epsilon_{23}$	0	$\epsilon_{22} \epsilon_{32}$	0
	0	0	$\epsilon_{23} \epsilon_{13}$	$\epsilon_{23} \epsilon_{21}$	$\epsilon_{23} \epsilon_{22}$	ϵ_{23}^2	0	0	$\epsilon_{23} \epsilon_{33}$
Firm 3	$\epsilon_{31} \epsilon_{11}$	0	0	$\epsilon_{31} \epsilon_{21}$	0	0	ϵ_{31}^2	$\epsilon_{31} \epsilon_{32}$	$\epsilon_{31} \epsilon_{33}$
	0	$\epsilon_{32} \epsilon_{12}$	0	0	$\epsilon_{32} \epsilon_{22}$	0	$\epsilon_{32} \epsilon_{31}$	ϵ_{32}^2	$\epsilon_{32} \epsilon_{33}$
	0	0	$\epsilon_{33} \epsilon_{13}$	0	0	$\epsilon_{33} \epsilon_{23}$	$\epsilon_{33} \epsilon_{31}$	$\epsilon_{33} \epsilon_{32}$	ϵ_{33}^2

Figure 6
Residual cross product matrix: Firm and time effects
This figure shows a sample covariance matrix of the residuals. When standard errors are clustered by both firm and time, residuals of the same firm in different years, as well as residuals of the same year but on different firms, may be nonzero. Observations on different firms and different years are assumed to be zero and are reported as zero in the matrix.

Figure 6: firm and time covariance matrix.

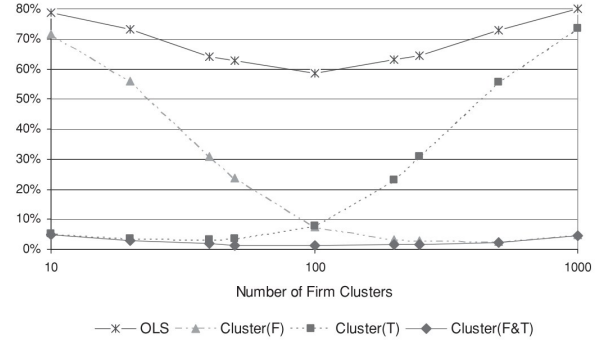


Figure 7
Rejection rates in the presence of a firm and a time effect for t -statistics clustered by firm, by time, or both
The figure graphs the fraction of t -statistics that are statistically significant at the 1% level (greater than 2.58 in absolute value). The number of firms and time periods range from 10 to 1,000 so that the total number of observations is always 10,000. The number of firms increases from 10 to 1,000 as we move left to right across the figure, while the number of time periods decreases from 1,000 to 10 as we move left to right across the figure. Thus the number of firms and the number of years in each of the nine simulations are (10, 1000), (20, 500), (40, 250), (50, 200), (100, 100), (200, 50), (250, 40), (500, 20), (1000, 10). In each simulation, OLS standard errors (stars), standard errors clustered by firm (triangles), standard errors clustered by year (squares), and standard errors clustered by firm and year (diamonds) are reported.

Figure 7: rejection rates under one- and two-way clustering.

5.6 Table 5 and Figure 8: temporary firm effects and adjusted Fama–MacBeth

Read:

- Table 5 studies cases in which the firm effect is persistent but not necessarily fixed.
- Panel A compares OLS, OLS with firm dummies, and firm-clustered standard errors.
- Panel B compares GLS standard errors with and without clustering.
- Panel C compares unadjusted and AR(1)-adjusted Fama–MacBeth standard errors.
- Figure 8 shows that adjusted Fama–MacBeth can help but does not generally fix all false positives.

Economic message: Fixed effects and GLS are not substitutes for valid standard errors. Even after modeling part of the dependence, inference can still be wrong unless the remaining correlation is handled.

Table 5
Estimated standard errors with a nonfixed firm effect

Panel A: OLS and clustered standard errors				
	I	II	III	IV
Avg($\hat{\rho}_{OLS}$)				
Std($\hat{\rho}_{OLS}$)				
Avg($\hat{\sigma}_{OLS}$)				
% Sig(T_{OLS})				
Avg($\hat{\sigma}_{OLS}$)				
% Sig(T_C)				
ρ_X/ρ_ϵ	0.50/0.50	0.00/0.00	0.25/0.25	0.60/0.35
ϕ_X/ϕ_ϵ	0.00/0.00	0.90/0.90	0.75/0.75	0.99/0.81
OLS	0.9994 0.0513 0.0283 [0.1578] 0.0508 [0.0114]	1.0001 0.0659 0.0283 [0.2746] 0.0668 [0.0086]	1.0009 0.0566 0.0283 [0.1996] 0.0569 [0.0104]	0.9991 0.0677 0.0253 [0.3302] 0.0670 [0.0098]
OLS with firm dummies	1.0007 0.0299 0.0298 [0.0096] 0.0298 [0.0100]	1.0003 0.0517 0.0298 [0.1382] 0.0516 [0.0098]	1.0013 0.0442 0.0298 [0.0802] 0.0441 [0.0092]	1.0046 0.1881 0.1101 [0.1288] 0.1886 [0.0108]
Panel B: GLS estimates with and without clustered standard errors				
	I	II	III	IV
Avg($\hat{\rho}_{GLS}$)				
Std($\hat{\rho}_{GLS}$)				
Avg($\hat{\sigma}_{GLS}$)				
% Sig(T_{GLS})				
Avg($\hat{\sigma}_{GLS-C}$)				
% Sig(T_{GLS-C})				
ρ_X/ρ_ϵ	0.50/0.50	0.00/0.00	0.25/0.25	0.60/0.35
ϕ_X/ϕ_ϵ	0.00/0.00	0.90/0.90	0.75/0.75	0.99/0.81
GLS	1.0005 0.0284 0.0283 [0.0090] 0.0282 [0.0090]	1.0003 0.0475 0.0283 [0.1240] 0.0474 [0.0100]	1.0012 0.0408 0.0283 [0.0730] 0.0408 [0.0090]	1.0006 0.0731 0.0580 [0.0388] 0.0721 [0.0112]

Table 5, panels A–B: nonfixed firm effects, OLS/GLS.

Table 5
(Continued)

Panel C: Fama-MacBeth standard errors				
	I	II	III	IV
Avg($\hat{\rho}_{FM}$)				
Std($\hat{\rho}_{FM}$)				
Avg($\hat{\sigma}_{FM}$)				
% Sig(T_{FM})				
Avg($\hat{\sigma}_{FM-AR(1)}$)				
% Sig($T_{FM-AR(1)}$)				
ρ_X/ρ_ϵ	0.50/0.50	0.00/0.00	0.25/0.25	0.60/0.35
ϕ_X/ϕ_ϵ	0.00/0.00	0.90/0.90	0.75/0.75	0.99/0.81
Fama-MacBeth	0.9995 0.0514 0.0239 [0.2510] 0.0224 [0.3222]	1.0001 0.0660 0.0187 [0.4696] 0.0389 [0.2876]	1.0008 0.0267 0.0221 [0.3350] 0.0376 [0.2998]	0.9991 0.0667 0.0138 [0.6094] 0.0289 [0.3900]
Avg(first-order autocorrelation)	-0.1157	0.4395	0.3250	0.4389

The table contains estimates of the coefficient and standard errors based on five thousand simulated panel data sets, each of which contains five hundred firms and ten years per firm. The true slope coefficient is 1, the standard deviation of the independent variable is 1, and the standard deviation of the error term is 2. Across the columns the magnitude of the fixed firm effect (ρ) and the first-order autocorrelation (ϕ) varies. ρ_X (ρ_ϵ) is the fraction of the independent variable's (residual's) variance that is due to the fixed firm effect. ϕ_X (ϕ_ϵ) is the first-order autocorrelation of the nonfixed portion of the firm effect of the independent variable (residual). Combining Equation (4) with Equation (18), the residual is specified as

$$\epsilon_{it} = \gamma_i + \eta_{it} = \gamma_i + \epsilon_{it} \quad \text{if } t = 1 \\ = \gamma_i + \phi \eta_{i,t-1} + \sqrt{1 - \phi^2} \epsilon_{it} \quad \text{if } t > 1.$$

The independent variable is specified in a similar manner.

Panel A contains coefficients estimated by OLS. In the first row, only the independent variable (X) is included; in the second row, firm dummies are also included in the regression. The first two entries in each cell contain the average slope estimated by OLS and the standard deviation of the coefficient (i.e., the true standard error). The third entry is the average standard error estimated by OLS. The percentage of OLS t -statistics that are significant at the 1% level (e.g., $|t| > 2.58$) is reported in square brackets. The fifth entry is the average standard error clustered by firm. The percentage of clustered t -statistics that are significant at the 1% level is reported in square brackets.

Panel B contains coefficients and standard errors estimates of the random effects model using feasible generalized least-squares (FGLS). The first two entries in each cell contain the average slope estimated by GLS and the standard deviation of the coefficient (i.e., the true standard error). The third entry is the regular standard error estimated by GLS (i.e., not clustered). The percentage of regular GLS t -statistics that are significant at the 1% level (e.g., $|t| > 2.58$) is reported in square brackets. The fifth entry is the average standard error clustered by firm. The percentage of clustered t -statistics that are significant at the 1% level is reported in square brackets.

Panel C contains coefficients and standard errors estimated by Fama-MacBeth. The first two entries in each cell contain the average slope estimated by Fama-MacBeth and the standard deviation of the coefficient (i.e., the true standard error). The third entry is the average standard error estimated by the Fama-MacBeth procedure (see Equation (9)). The percentage of t -statistics that are significant at the 1% level (e.g., $|t| > 2.58$) is reported in square brackets. The fifth and sixth entries are the Fama-MacBeth standard error corrected for the first-order autocorrelation and the percentage of t -statistics that are significant at the 1% level. I adjust the standard error by multiplying the traditional Fama-MacBeth standard error (Equation (9)) by the square root of $(1 + \rho(1 - \theta))$, where θ is the first-order autocorrelation of β and β_{t-1} . The average first-order autocorrelation is reported in the last row of panel C.

Table 5, panel C: Fama-MacBeth and AR(1) adjustment.

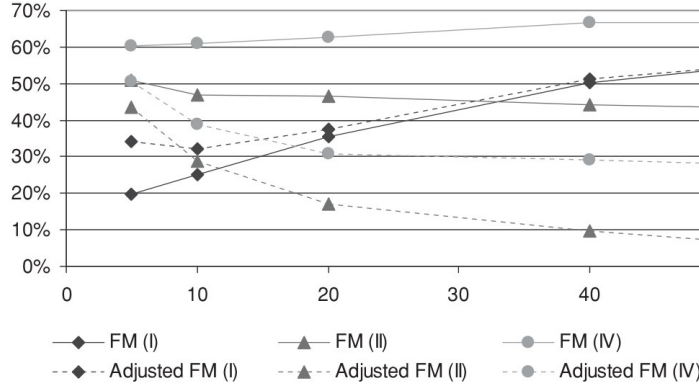


Figure 8

Rejection rates based on unadjusted and adjusted Fama-MacBeth standard errors

The figure graphs the fraction of t -statistics that are statistically significant at the 1% level (greater than 2.58 in absolute value) as a function of number of time periods per firm. The rejection rates are graphed for both standard (unadjusted) Fama-MacBeth t -statistics and Fama-MacBeth t -statistics adjusted for the first-order serial correlation. The rejection rates are based on three data structures from Table 5. The rejection rates for data with only a fixed firm effect ($\rho = 0.50$ and $\phi = 0.00$, column I in Table 5) are graphed with diamonds. Rejection rates for data with only a first-order autocorrelation ($\rho = 0.00$ and $\phi = 0.90$, column II in Table 5) are graphed with triangles. A data structure with both a fixed firm effect and an autoregressive firm effect ($\rho_X = 0.60$, $\phi_X = 0.99$, $\rho_\epsilon = 0.35$, $\phi_\epsilon = 0.81$, column IV in Table 5) is graphed with circles. The symbols with solid lines are the rejection rates based on unadjusted Fama-MacBeth t -statistics, while the symbols with dotted lines are the rejection rates based on adjusted Fama-MacBeth t -statistics.

5.7 Table 6 and Figure 9: asset-pricing application

Read:

- Table 6 reproduces an equity-return regression with asset tangibility and related characteristics.
- White and firm-clustered standard errors are similar because residual autocorrelation within firm is effectively zero.
- Time-clustered and two-way-clustered standard errors are much larger, indicating strong cross-sectional dependence within months.

- Figure 9 shows that residuals and independent variables have little within-firm residual autocorrelation but substantial within-month correlation.

Economic message: In asset pricing, common shocks often dominate firm persistence. Time clustering can be much more important than firm clustering.

Table 6
Asset pricing application: Equity returns and asset tangibility

	I	II	III	IV	V
$\text{Log}(B/M)_{t-5}$	0.2456** (0.0238)	0.2456** (0.0247)	0.2456** (0.0924)	0.2456** (0.0926)	0.2064** (0.0795)
$\text{Log}(\text{Book Return})$ (last five years)	0.2482** (0.0385)	0.2482** (0.0396)	0.2482** (0.0859)	0.2482** (0.0864)	0.2145** (0.0788)
Market return (last five years)	-0.3445* (0.0257)	-0.3445* (0.0261)	-0.3445* (0.1000)	-0.3445* (0.1001)	-0.3310** (0.0893)
Share issuance	-0.5245** (0.0426)	-0.5245** (0.0427)	-0.5245** (0.1440)	-0.5245** (0.1441)	-0.5143** (0.1235)
R^2	0.0008	0.0008	0.0008	0.0008	0.0008
Coefficient estimates	OLS	OLS	OLS	OLS	FM
Standard errors	White	CL - F	CL - T	CL - F&T	FM

The table contains coefficient and standard error estimates of the equity return regressions from Daniel and Titman (2006). The data are briefly described in the Appendix and in detail in their paper. The sample runs from July 1968 to December 2003 and contains 844,155 firm-month observations. The estimates in columns I-IV are OLS coefficients and the regressions contain time (month) dummies. Standard errors are reported in parentheses. White standard errors are reported in column I, standard errors clustered by firm in column II, by month in column III, and by firm and month in column IV. Column V contains coefficients and standard errors estimated by Fama-MacBeth. Statistical significance at the 1% and 5% levels are denoted by ** and *, respectively.

Table 6: equity returns and asset tangibility.

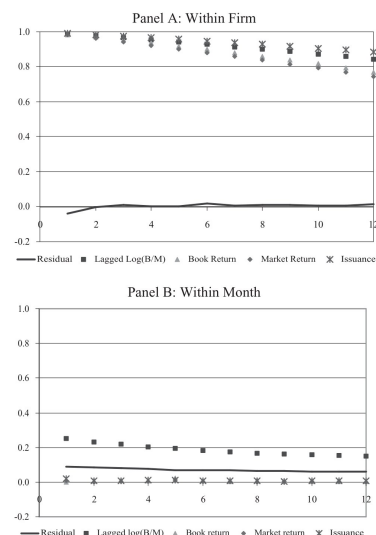


Figure 9
Residuals and independent variables autocorrelation: asset pricing example
The autocorrelations of the residual and the four independent variables are graphed for one to twelve lags. In panel A, the correlations are within firm and are only calculated for observations of the same firm [i.e., $\text{Corr}(e_{it}, e_{it-k})$ for k equal 1 to 12]. In panel B, the correlations are within month and are only calculated for observations in the same month [i.e., $\text{Corr}(e_{it}, e_{it-k})$ for k equal 1 to 12]. The data were sorted by month and then industry (four-digit SIC code) in panel B. The independent variables are described in the Appendix.

Figure 9: autocorrelation in asset-pricing data.

5.8 Table 7 and Figure 10: corporate-finance application

Read:

- Table 7 estimates a capital-structure regression using annual firm-level data from 1965–2003.
- Firm-clustered standard errors are three to four times larger than White standard errors for many variables.
- Time-clustered standard errors are close to White standard errors, so time effects are less important in this application.
- Figure 10 shows strong within-firm persistence in the residual and regressors.

Economic message: Corporate-finance panel regressions often have strong firm effects because leverage, profitability, assets, and residual components persist across years. Firm clustering is therefore essential.

Table 7
Corporate finance application: Capital structure regressions (1965–2003)

	I	II	III	IV	V
Ln(MV assets)	0.0460** (0.0055)	0.0460* (0.0184)	0.0460** (0.0074)	0.0460* (0.0191)	0.0394** (0.0076)
Ln(1 + Firm age)	−0.0432** (0.0084)	−0.0432 (0.0297)	−0.0432** (0.0067)	−0.0432 (0.0293)	−0.0479** (0.0077)
Profits/sales	−0.0330** (0.0107)	−0.0330 (0.0359)	−0.0330** (0.0098)	−0.0330 (0.0357)	−0.0299** (0.0097)
Tangible assets	0.1043** (0.0057)	0.1043** (0.0197)	0.1043** (0.0083)	0.1043** (0.0206)	0.1158** (0.0096)
Market-to-book (assets)	−0.0251** (0.0006)	−0.0251** (0.0020)	−0.0251** (0.0013)	−0.0251** (0.0023)	−0.0272** (0.0016)
Advertising/sales	−0.3245** (0.0841)	−0.3245 (0.2617)	−0.3245** (0.0814)	−0.3245 (0.2609)	−0.3965* (0.1712)
R&D/sales	−0.3513** (0.0469)	−0.3513* (0.1544)	−0.3513** (0.0504)	−0.3513* (0.1555)	−0.3359** (0.0501)
R&D > 0 (= 1 if yes)	0.0177** (0.0024)	0.0177* (0.0076)	0.0177** (0.0025)	0.0177* (0.0077)	0.0126** (0.0034)
R-squared	0.1360	0.1360	0.1360	0.1360	0.1300
Coefficient estimates	OLS	OLS	OLS	OLS	FM
Standard errors	White	CL – F	CL – T	CL – F&T	FM

The table contains coefficient and standard error estimates of a capital structure regression. The dependent variable is the market debt ratio (book value of debt divided by the sum of the book value of assets minus the book value of equity plus the market value of equity). The data are annual observations between 1965 and 2003. The sample contains NYSE firms that pay a dividend in the previous year. There are 24,286 firm-years in the sample. The independent variables are lagged one year and are defined in the Appendix. The estimates in columns I–IV are OLS coefficients and the regressions contain time (year) dummies. Standard errors are reported in parentheses. White standard errors are reported in column I, standard errors clustered by firm in column II, by month in column III, and by firm and month in column IV. Column V contains coefficients and standard errors estimated by Fama–MacBeth. Statistical significance at the 1% and 5% levels are denoted by ** and *, respectively.

Table 7: capital structure regression.

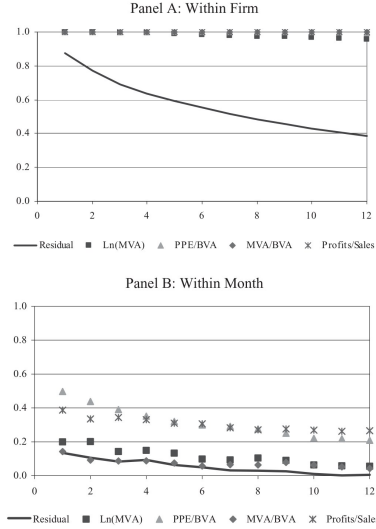


Figure 10
Residuals and independent variables autocorrelation: Corporate finance example
The autocorrelations of the residual and four of the right independent variables are graphed for one to twelve lags. In panel A, the correlations are within firm and are only calculated for observations of the same firm (i.e., $\text{Corr}(y_t, y_{t-k})$ for k equal 1 to 12). In panel B, the correlations are within year and are only calculated for observations of the same year (i.e., $\text{Corr}(y_t, y_{t-k})$ for k equal 1 to 12). The data were sorted by month and then industry (four-digit SIC code) in panel B. The independent variables are described in the Appendix. The graphs for the remaining four variables are similar and are available from the author.

Figure 10: autocorrelation in corporate-finance data.

6 Short summary

- Petersen studies standard-error estimation in finance panel data.
- OLS and White standard errors are correct only when residuals are independent across observations.
- A firm effect creates within-firm dependence; standard errors should be clustered by firm.
- A time effect creates cross-sectional dependence within periods; standard errors should be clustered by time or estimated using Fama–MacBeth.
- If both firm and time effects matter, two-way clustering is the appropriate default.
- Bias depends on whether both the regressor and residual are correlated in the same dimension.
- Fama–MacBeth does not fix firm effects, and Newey–West does not generally substitute for firm clustering.
- Fixed effects and GLS may improve coefficient estimation but do not automatically fix inference.
- Few clusters are a problem even for clustered standard errors.
- Comparing White, firm-clustered, time-clustered, and two-way-clustered standard errors is a useful diagnostic.
- The asset-pricing example is dominated by time effects; the corporate-finance example is dominated by firm effects.

7 Conclusion

7.1 Core conclusion

Petersen’s central conclusion is that no single standard-error method is universally correct for finance panel data. The correct method depends on the structure of residual dependence. Researchers should cluster in the dimension in which residuals and regressors are dependent, and cluster in two dimensions when both firm and time dependence matter.

7.2 Economic intuition

Panel observations are not independent simply because they are numerous. If a firm characteristic persists across years, repeated observations from the same firm contain less independent information than OLS assumes. If a market-wide shock hits many firms in the same month, cross-sectional observations also contain less independent information. Standard errors must be adjusted to count information, not just observations.

7.3 Takeaway

The practical rule is: diagnose the dependence structure and match the standard-error estimator to it. In many corporate-finance settings, cluster by firm. In many asset-pricing settings, cluster by time or use Fama–MacBeth. When both dimensions are plausible, report two-way clustered standard errors and compare them with one-way alternatives.